

ALIAH UNIVERSITY
End-Term Examination- 7th Semester
Session-2023-2024

Year: 4th Year
Semester: 7th Semester
Subject: Irrigation Engineering
Subject Code: CENUGPE04

Full Marks: 80
Time: 3hr.

Answer question no.1 and any three from the rest:

1. Write short notes on the following:

[4*5=20]

- i) Aqueduct and syphon aqueduct.
- ii) Border strips and sprinkler methods of irrigation fields.
- iii) Water-shed canals and contour canals.
- iv) Culturable command area and intensity of irrigation.

2. a) What is meant by 'Duty' and 'Delta' of canal water? Derive a relationship between duty and delta for a given base period.

[4+4=8]

b) Explain the influence of several factors which affect duty.

[6]

c) Water is released at the rate of 5 cumecs at the head sluice. If the duty at the field is 100 hectares/cumec and the loss of water in transit is 30%, find the area of the land that can be irrigated.

[6]

3. a) What are the adverse effect of silting and scouring in irrigation canals?

b) Define: i) True regime channel ii) Critical velocity in channel according to Kennedy.

c) Design a regime trapezoidal channel having side slope 0.5H:1V for a discharge of 50 cumecs, silt factor=1.1, using Lacey's theory.

[5+5+10=20]

4. a) What are the main types of channel linings?

b) Differentiate between a weir and a barrage.

c) Design a lined canal to carry a discharge of 15 cumec. The available and acceptable country side slope is 1 in 9000. Assume, suitable values of side slope and good brick work in lining ("f" value for brick=0.015).

[5+5+10=20]

5. a) What do you mean by diversion head work?

[2]

b) Draw a neat layout of diversion headworks indicating various components of the system. Briefly write the function of each component.

[8]

c) i) What is meant by a "Cross-Drainage Works"? ii) Explain- why such works are not required within a ridge canal system. iii) What are the different types of cross drainage works that are necessary on a canal alignment? State briefly the conditions under which each one is used.

[2+2+6=10]

Course: Civil Engineering

Year: Final Year

Subject: Air and Noise Pollution and Control

Subject Code: (CENUGPE03) (PEC III)

Max. Marks: 80

Time: 3 hrs

1. Answer all questions: (25X2=50)

- a) Derive the expressions for sound intensity level, sound pressure level and sound power level. Establish the relationship between the three. Derive the basic wave equation (9+8+8=25)
- b) Enumerate the basic Gaussian Plume Dispersion Equation clearly defining all the terms involved in the expression. Modify the equation as per the following conditions: i) the receptor is at ground level with no ground reflection having taken place ii) both the source and the receptor are at ground level with no ground reflection iii) plume centreline ground level concentration with no ground reflection. What are the assumptions behind the basic Gaussian Plume Dispersion Model? Derive the followings i) $\sigma_z = 0.707H$ and $\sigma_z = 0.47H$, where the notations have their usual meanings. (4+6+3+6+6=25)

2. Answer all questions: (15X2=30)

- i) It is estimated that 150 gm/sec of sulphur dioxide is being emitted from a petroleum refinery from an average effective stack height of 150 m. At 9 am on an overcast winter morning with a surface wind speed of 9 m/sec, what will be the ground level concentration directly downwind from the refinery at a distance of 800 m. Assume $\sigma_y = 40m$ and $\sigma_z = 20m$. (7.5)
- ii) A burning dump emits 5.5 gm/sec of oxides of nitrogen. Find out the concentration of oxides of nitrogen averaged over approximately 10 mins, from this source directly downwind at a distance of 4.5 km on an overcast night with a wind speed of 6 m/sec. Consider the dump to be a point ground-level source with no effective rise. Assume $\sigma_y = 220m$ and $\sigma_z = 80m$ respectively. (7.5)

b) Traffic Noise data are shown in the table below: ((15)

Time (s)	Sound Pressure Level.dB(A)
10	70
20	75
30	77
40	69
50	83
60	86
70	79
80	74
90	71
100	72

Compute L_{eq} .

Odd (Autumn) Semester Examination 2023

Paper Code: MBAUGHU01; Paper name: Industrial Economics and Management

MEN/EEN/CEN/CSE/ECE VIIth Semester

Full Marks: 80; Time: 3Hrs.

(The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as possible)

GROUP: A (Answer all the questions)

(1 x 10 = 10)

1. I. A situation, wherein a perceiver tends to see in others the traits he himself possesses, is called _____.

- (a) Repetition ~~(b) Contrast~~ (c) Rejection ~~(d) Projection~~

II. A written summary of content & context of job is called _____.

- ~~(a) Job description~~ (b) Job specification (c) Resume (d) Job posting

III. Inspection, scrap, and repair are examples of _____.

- ~~(a) Internal costs~~ (b) External costs (c) Societal costs (d) Costs of dissatisfaction

IV. OB helps to understand behaviour of human in _____.

- (a) Work place only (b) Society only (c) Department only ~~(d) Work place and Society~~

V. Which of the following could be a strength?

- (a) Weather ~~(e) A new international market~~
~~(b) A price that is too high~~ ~~(d) The location of a business~~

VI. Which of the following is the relation that the law of demand defines?

- (a) Income and price of a commodity (c) Price and quantity of a commodity
(b) Income and quantity demanded (d) Quantity demanded and quantity supplied

VII. Numerous forms of short-term incentives to promote trial or buying of a service or product is

- (a) Sales promotion (b) Direct marketing (c) Advertising ~~(d) Events and experiences~~

VIII. _____ is the task of buying goods of the right quality, in the right quantities, at the right time and at the right price.

- ~~(a) Supplying~~ (b) Purchasing (c) Scrutinizing (d) None of the above

IX. _____ is the set of forces that energize, direct, and sustain behaviour.

- ~~(a) Motivation~~ (b) Expectancy (c) Empowerment (d) Socialization

X. Raju believes that men perform better in oral presentations than women. What shortcut has been used in this case?

- (a) The halo effect (b) Projection ~~(c) The contrast effect~~ (d) Stereotyping

GROUP: B (Answer any five questions)

(5 x 5 = 25)

2. What is quality management? Describe the Cycle of 'TQM (Total Quality Management)'.
3. Describe the process of communication.
4. Elaborate the importance of economics in business world.
5. What do you mean by marketing research? What are the various steps in the marketing research process?
6. What is elasticity of demand? Consider the demand for a good. At price Rs 4, the demand for the good is 25 units. Suppose price of the good increases to Rs 5, and as a result, the demand for the good falls to 20 units. Calculate the price elasticity?

7. What is the importance of training & development?
8. What do you mean by the term performance appraisal? What are the advantages and disadvantages of performance appraisal?

GROUP: C (Answer any three questions) (15 x 3 = 45)

9. (a) What is marketing mix? Describe 4P's of marketing. (1+4)
(b) Describe segmentation, targeting and positioning (STP) with the help of an example. (10)
10. What do you mean by law of demand? What are the factors that affect demand? What are the exceptions to law of demand? (2+5+8)
11. What are the various types of barriers to effective communication? Describe with the help of examples. (1+4+10)
12. What is motivation? What is the process of motivation? Explain Maslow's need hierarchy theory and its limitation. (3*5=15)
13. Differentiate between the following terms:
 - i. Training and Development
 - ii. Maslow's need hierarchy theory and Herzberg's two-factor theory
 - iii. Projection and Stereotyping
 - iv. Complementary and Supplementary Goods
 - v. Selling and Marketing

Aliah University

Odd (Autumn) Semester Examination, 2023

Paper Code: MBAUGOE01; Paper name: Entrepreneurship Development
B.Tech VIIth Semester (CEN/MEN/EEN)

Full Marks: 80; Time: 3Hrs.

(The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as possible)

GROUP: A (Answer all the questions) (1 x 10 = 10)

1.

I. According to Schumpeter, innovative entrepreneurs would:

- a. Thrive in the market
- b. Not survive and disappear from the market.
- c. Get absorbed within larger innovative businesses
- d. Get absorbed within non-innovative businesses

II. Which of these is not a challenge for the entrepreneur?

- a. Managing the cash flow of their business
- b. Recruiting new employees
- c. Formulating rules and regulations relating to conducting entrepreneurship in their country
- d. Choosing the product or service to sell in the market

III. Which of these theories involve taking a moderate amount of risk as a function of skill and not chance?

- a. Need for achievement
- b. Need for authority
- c. Need for affiliation
- d. Need for independence

IV. Which of these is not a category of external forces that affects a business?

- a. Competitive forces
- b. Economic forces
- c. Technological forces
- d. Socio-economic forces

V. A business arrangement where one party allows another party to use a business name and sell its products or services is known as _____.

- a. a cooperative
- b. an owner-manager business
- c. a franchise
- d. a limited company

VI. Primary data is _____.

- a. the most important data
- b. the data that is collected from news paper
- c. new data specifically collected for a project
- d. data that is collected second.

VII. Goods or services reach the market place through _____.

- a. monopolies
- b. multilevel pyramids
- c. marketing channels
- d. multiplication.

VIII. The application for registration of a small scale unit should be submitted to _____

- a. General Manager, NSIC
- b. Director, DIC
- c. General Manager, DIC
- d. Director, NSIC

IX. In terms of innovative business the five policy includes _____

- a. Equity shares, private sources, preference shares, debentures and capital investment
- b. Equity shares, Preference shares, debentures, mutual fund and Insurance
- c. Private sources of funding, Debt financing, Venture capital, Business angels, subsidies and grants from governments
- d. None of these

X. Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE) was set up by Government of India and which of the following bank?

- a. RBI b. NABARD c. SIDBI d. MUDRA

GROUP: B (Answer any five questions)

(5 x 5 = 25)

- 2. Who is an Entrepreneur? What are the characteristics of Entrepreneurs?
- 3. What are the types of Entrepreneurs?
- 4. Discuss 'Project Life Cycle' in brief.
- 5. What is the difference between Entrepreneur vs. Intrapreneur?
- 6. Point out few drawbacks of Entrepreneurship.
- 7. Write a short note on 'Small Scale Industries'.
- 8. Highlight the problems associated with 'Start-up'.

GROUP: C (Answer any three questions)

(15 x 3 = 45)

- 9. Discuss the role of Entrepreneurship in economic development.
- 10. Elucidate in brief the problems faced by Women Entrepreneurs.

11. Write short notes on: (3x5)

- (i) Sole Proprietorship
- (ii) Joint Stock Company
- (iii) Cooperative Organisation

- 12. (a) What do you understand by 'Project Identification'?
- (b) Discuss the steps in 'Project Identification'.
- (c) Elaborate about 'Project Classification'.

(4+5+6)

13. Cite the example of two institutions/ organizations which support the entrepreneurs for their entrepreneurial activities. Discuss the details of those two institutions/ organizations. (3+12)

OR

Write a business plan of your choice, with suitable illustration.
